MIL-S-48735A(AR)
AMENDMENT 4
7 February 1986
SUPERSEDING
AMENDMENT 3
22 October 1984

MILITARY SPECIFICATION

SAFE AND ARMING DEVICE FOR MINES, M718/M741
SYSTEMS METAL PARTS FOR

This Amendment forms a part of Military Specification MIL-S-48735A(AR), dated 31 March 1982, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1: Add the following under SPECIFICATIONS; MILITARY:

"MIL-P-50884 - Printed Wiring, Flexible, General Specification For.

MIL-S-45743 - Soldering, Manual type, High Reliability Electrical and Electronic Equipment."

Add the the following under STANDARDS; MILITARY:

"MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts."

2.1.2: Delete drawing "9278001 - Clip, Grounding" and add drawing "9317651 - Detonator Grounding Clip".

PAGE 3

2.1.2: Add to American Society for Testing and Materials:

"ASTM B 487-79 - Measurement of Metal and Oxide Coating Thickness by Microscopic Examination of a Cross Section."

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* "ASTM B525-83 - Sintered Austenitic Stainless Steel Structural Parts."
* "ASTM B328-73 - Density & Interconnected Porosity of Sintered Powder Metal Structural Parts & Oil-Impregnated Bearings."

PAGE 3

3.6.5: Delete paragraph in its entirety.

3.7: Delete paragraph in its entirety.

3.8: Add new paragraph as follows:

"3.8 Flexcable solder pads.

3.8.1 Plating. The flexcable shall have a uniformity in plating on all solder pads. The plating shall contain no foreign material. There shall be no evidence of any lifting or separation of plating from the surface of the pad. The coverlayer access hole registration shall be such that the diameter of the access hole, including extruded adhesive, shall not reduce the solder ring areas below the limits specified on the drawing.

3.8.2 Lifted pads. The flexcable shall contain no lifted pads.

3.8.3 Solderability. When the flexcable is tested for solderability in accordance with MIL-P-50884, the flexcable shall show full wetting and new solder coverage of at least 95% of conductor area not covered by insulating material or extruded adhesive. There shall be no distortion or delamination."

PAGE 4

4.3.1: Delete "Clip, Grounding 9278001 10" and Add "Detonator Grounding Clip 9317651 10".

PAGE 5

TABLE I: Delete in its entirety and replace with new page 5 included.

PAGE 6

TABLE I: Delete in its entirety and replace with new page 6 included.

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PAGE 8

4.4.2.1: Delete in its entirety and replace with new 4.4.2.1 included.

PAGE 27

* 4.4.2.20 Dual MDF Chamber.

Major		
106	Change "holes (2)" to "hole"	
107	Change "holes" to "hole"	
Minor		
203	Change "hole(s)" to "hole"	

Add "Major 111 Material Improper, -,, 3.1, Certification. Major 112 Density Improper, 30, 3.1, 4.4.5/4.5.8"

* Add note 1:

"The inspections and tests of Major 111 and Major 112 are applicable when using alternate material, Sintered Austenitic Stainless Steel."

PAGE 30

4.4.2.23: Delete in its entirety and replace with new page 30 included.

PAGES 32 - 36

4.4.2.25: Delete in its entirety and replace with new pages 32-36b included.

PAGE 43

4.4.2.31: Retainer, spring. Add the following Major Defects:

"104	O.D. of pilot	0.40%	3.2	Gage
105	True position of pilot	0.40%	3.2	Gage
106	Height of pilot	0.40%	3.2	Gage"

PAGE 56

* Add Para 4.4.5 Density. A sample of 30 Dual MDF Chambers shall be selected from each lot (batch) and tested for the specified density in accordance with 4.5.8."

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PAGE 57

4.5.7: Add new paragraph as follows:

"4.5.7: Flexcable solder pads.

4.5.7.1: Plating. The flexcable solder pads shall be examined under a stereo microscope with ten power to thirty power (10X to 30X) magnification. Lighting shall be in accordance with inspection provisions as required in MIL-S-45743. If a solder pad is suspect as being non-solderable it shall be tested for solderability by applying heat from a twenty-five watt (25W) soldering iron to the pad and adding cored solder with only RMA flux to the heated junction. If the solder fails to fully wet the total pad area within a maximum of three (3) seconds, the pad shall be considered as non-solderable. Any pad failing to comply with all of the specified requirements shall cause the cable to be classed defective. Any cable classed defective shall be removed from the lot. (Non-destructive test).

4.5.7.2: Solderability. All pads on the flexcable shall demonstrate the wetting and solderability requirements over all specified pad surface area when tested in accordance with MIL-P-50884. The only pad areas that may be free of solder wetting are those specified in the requirements.

Any pad failing to comply with all of the specified requirements shall cause the flexcable to be classed defective. Samples undergoing testing shall not be returned to the lot."

* Add "Para 4.5.8 Density. The Dual MDF chamber shall be tested for density iln accordance with ASTM Method B328-73 (or equivalent). Any Dual MDF chamber failing to comply with the specified requirements shall be classified defective and shall cause the lot (batch) represented by the test sample to be rejected."

PAGE 58

* 6.2 Delete "DRSMC-QAT-I(D)" and substitute "AMSMC-QAT-I(D)"

* 6.3 Delete "DRSMC-QAT-M(D)" and substitute "AMSMC-QAT-M(D)"

6.4 Delete "ARRADCOM" and substitute "AMCCOM"

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The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	S & A Metal Parts				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cable, Flexible (Dwg. 9277774)				
	Examination for defects	10		3.2	4.4.2.1
	Continuity resistance	10		3.4	4.5.4
	Plating	25		3.8.1	4.5.7.1
	Lifted pads	25		3.8.2	4.4.2.1
	Solderability	25		3.8.3	4.5.7.2
	Lock, Centrifugal (Dwg. 9277786)				
	Examination for defects	10		3.2	4.4.2.2
	"G" Weight (Dwg. 9277790)				
	Examination for defects	10		3.2	4.4.2.3
	Hardness	Note 1		3.2	4.5.1

NOTES: 1. Ten (10) parts shall be selected from each heat treat batch.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: S & A Metal Parts		SHEET 2 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY: --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Detonator Grounding Clip (Dwg. 9317651)				
	Examination for defects	10		3.2	4.4.2.23
	Solderability	5		3.2	4.4.2.23
	Plating thickness	5		3.2	4.4.2.23
	Housing (Dwg. 9278008)				
	Examination for defects	10		3.2	4.4.2.25
	Slider (Dwg. 9278086)				
	Examination for defects	10		3.2	4.4.2.29
	Hardness	Note 1		3.2	4.5.1
	Setback Pin (Dwg. 9317584)				
	Examination for defects	10		3.2	4.4.2.34
	Hardness	Note 1		3.2	4.5.1

NOTES: 1. Ten (10) parts shall be selected from each heat treat batch.

7(6)

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cable, Flexible	1 OF 1	9277774
			NEXT HIGHER ASSEMBLY 9277868

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of terminal holes		0.40%	3.2	Gage
102	Diameter of terminal holes		0.40%	3.2	Gage
103	Continuity resistance		0.40%	3.4	4.5.4
104	Any bridging of copper between conductive paths		0.40%	3.2	Visual
105	Any separation of plastic lamination		0.40%	3.2	Visual
106	Lifted pads		0.40%	3.8.2	Visual
107	Solderability (Note 1)	25-0-1		3.8.3	4.5.7.2
108	Plating		0.40%	3.8.1	4.5.7.1
Minor					
201	Profile of cable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: 1. Sample 25 from each inspection lot. Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Detonator Grounding Clip	1 OF 1	9317651
			NEXT HIGHER ASSEMBLY --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of split ring (2 places)		0.40%	3.2	Gage
102	Length (.553-.005)		0.40%	3.2	Gage
103	Height of split ring (2 places)		0.40%	3.2	Gage
104	Depth to bottom of split ring (2 places)		0.40%	3.2	Gage
105	Distance from center line to edge of tab (.188-.005)		0.40%	3.2	Gage
106	Distance from split ring to edge of tab (.170-.005)		0.40%	3.2	Gage
107	Presence of chamfer		0.40%	3.2	Visual
108	Solderability of tab	25-0-1		3.2	Note 1
109	Plating thickness	5 0-1		3.2	Note 2
110	Plating improper		0.40%	3.2	Visual/Certificate of compliance
111	Material/Grain direction			3.1	Certificate of compliance
Minor					
201	Poor workmanship		0.65%	3.6	Visual
NOTES:	1. Sample 25 from each inspection lot.	Test	per MIL-STD-202,		Method 208, Reject
	2. Sample 5 from each inspection lot. examine split ring section on both sides defective.	Test	per ASTM of clip.	B487-79, Reject	ection sample and for one (1) or more

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 7	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)		9278008
			NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large outside diameter		0.40%	3.2	Gage
102	Small outside diameter		0.40%	3.2	Gage
103	True position of small O.D. to large O.D.		0.40%	3.2	Gage
104	Distance of small O.D. to flat		0.40%	3.2	Gage
105	Overall length		0.40%	3.2	Gage
106	Length from closed end to edge of shoulder		0.40%	3.2	Gage
107	Diameter of large center hole		0.40%	3.2	Gage
108	True position of large center hole		0.40%	3.2	Gage
109	Diameter of small center hole		0.40%	3.2	Gage
110	True position of small diameter center hole		0.40%	3.2	Gage
111	Diameter of counterbore at the small center hole end		0.40%	3.2	Gage
112	Depth of counterbore at the small center hole end		0.40%	3.2	Gage
113	True position of counterbore at the small center hole end		0.40%	3.2	Gage
114	Length from bottom of large diameter center hole to end of small diameter center hole		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 7	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)		9278008
			NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Radius of curvature from large diameter to small diameter center hole		0.40%	3.2	Gage
116	Diameter of long hole parallel to center hole		0.40%	3.2	Gage
117	True position of long hole parallel to center hole		0.40%	3.2	Gage
118	True position of hole perpendicular to long hole and center hole (SHT 2, center view, .159 + .005 - .001 hole)		0.40%	3.2	Gage
119	Diameter of hole perpendicular to long hole		0.40%	3.2	Gage
120	True position of small diameter hole (.064 + .002 hole) SHT 1, center view		0.40%	3.2	Gage
121	Diameter of small diameter hole (.064 + .002)		0.40%	3.2	Gage
122	True position of larger diameter hole, (.125 + .005 hole)		0.40%	3.2	Gage
123	True position of the four equally spaced holes		0.40%	3.2	Gage
124	Small diameter of the four equally spaced holes		0.40%	3.2	Gage
125	Large diameter of the four equally spaced holes		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 7	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)		9278008
			NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
126	Depth to shoulder of the four equally spaced holes		0.40%	3.2	Gage
127	Deleted				
128	Deleted				
129	Deleted				
130	Deleted				
131	True position of the two centrifugal lock holes (SH 1, center view, .250 + .003 holes)		0.40%	3.2	Gage
132	Large diameter of the two centrifugal lock holes		0.40%	3.2	Gage
133	Small diameter of the two centrifugal lock holes		0.40%	3.2	Gage
134	Depth of large diameter of centrifugal lock holes		0.40%	3.2	Gage
135	Diameter of C bore of centrifugal lock holes		0.40%	3.2	Gage
136	Depth of C bore of centrifugal lock holes		0.40%	3.2	Gage
137	True position of the long transverse hole (SHT 1, right view, .109 + 002 hole)		0.40%	3.2	Gage
138	Diameter of the transverse hole		0.40%	3.2	Gage
139	Diameter of the counterbore of transverse hole (2 sides)		0.40%	3.2	Gage
140	Depth of the counterbore of transverse hole (2 sides)		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 7		DRAWING NUMBER
4.4.2.25	Housing (Rear lock)				9278008
					NEXT HIGHER ASSEMBLY
					9278015
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
141	True position of large diameter transverse hole in slot		0.40%	3.2	Gage
142	Diameter of large transverse hole in slot		0.40%	3.2	Gage
143	Depth from bottom of transverse hole to O.D. of center hole (SHT 2, Sect LL, .028-.003 dimension)		0.40%	3.2	Gage
144	True position of small diameter hole in bottom of large diameter transverse hole		0.40%	3.2	Gage
145	Diameter of small diameter hole in bottom of large transverse hole		0.40%	3.2	Gage
146	Length of slot in bottom of large diameter transverse hole		0.40%	3.2	Gage
147	Width of slot in bottom of large diameter transverse hole		0.40%	3.2	Gage
148	True position of slot in bottom of large diameter transverse hole		0.40%	3.2	Gage
149	Diameter of transverse hole, Sect KK (.189 + .002 hole)		0.40%	3.2	Gage
150	True position of above transverse hole		0.40%	3.2	Gage
151	Location of hole perpendicular to transverse hole in sect KK at edge of small O.D.		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)	5 OF 7	9278008
			NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
152	Large diameter of hole perpendicular to transverse hole in section KK at edge of small O.D. (SHT 1, right view, .144 + .002 hole)		0.40%	3.2	Gage
153	True position of above hole		0.40%	3.2	Gage
154	Small diameter of hole perpendicular to transverse hole in sect KK at edge of smal .D. (SHT 1, right view, .067 + .004 - .001 hole)		0.40%	3.2	Gage
155	True position of above hole		0.40%	3.2	Gage
156	Total length of above hole		0.40%	3.2	Gage
157	Length of small diameter section of above hole		0.40%	3.2	Gage
158	Diameter of C bore at end center hole		0.40%	3.2	Gage
159	Depth of C bore of above hole		0.40%	3.2	Gage
160	Diameter of C bore at end hole paralleling center hole		0.40%	3.2	Gage
161	Depth of C bore of above hole		0.40%	3.2	Gage
162	Length of large O.D.		0.40%	3.2	Gage
163	Surface finishes improper		0.40%	3.2	Gage
164	Material improper		0.40%	3.2	Gage
165	Width of slot (.125 + .005 dimen.)		0.40%	3.2	Gage
166	Deleted				
167	Length of slot (Sht 2, Center view .395 - .005 dimen)		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 6 OF 7	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)		9278008
			NEXT HIGHER ASSEMBLY 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
168	Height from Top of Housing to edge of above slot		0.40%	3.2	Gage
169	Width of above slot		0.40%	3.2	Gage
170	Depth from small outside diameter to bottom of side cutout containing the two thru holes (.068 + .010 dimen.)		0.40%	3.2	Gage
171	Location of the two (2) detonator holes in the side cutout (.105 + .002 diameter holes, SHT 2, detail P)		0.40%	3.2	Gage
172	Diameter of the detonator holes (2 places) (.105 + .002 holes)		0.40%	3.2	Gage
173	True position of the above holes		0.40%	3.2	Gage
174	Diameter of the detonator grounding clip holes (2 places, .142 + .003 holes)		0.40%	3.2	Gage
175	True position of the above holes (2 places)		0.40%	3.2	Gage
176	Diameter of detonator output holes (2 places, .089 + .004 [illegible]les)		0.40%	3.2	Gage
177	True position of the above holes (2 places)		0.40%	3.2	Gage
178	Height from the shoulder of the detonator grounding clip holes to the top of the holes (2 places, .125 - .005 dimen.)		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Housing (Rear lock)	7 OF 7	9278008
			NEXT HIGHER ASSEMBLY: 9278015

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Angle of the shoulder between the small O.D. and large O.D.		0.65%	3.2	Gage
202	Angle of the slot		0.65	3.2	Gage
203	Depth of the long hole parallel to the center hole		0.65%	3.2	Gage
204	Depth to the second shoulder of the locking pin hole (SHT 1, center view, .30 + .01 dimen.)		0.65%	3.2	Gage
205	Deleted				
206	Evidence of poor workmanship		0.65%	3.6	Visual
207	Protective finish missing or improper		0.65%	3.2	Visual

NOTES:

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Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A359)

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